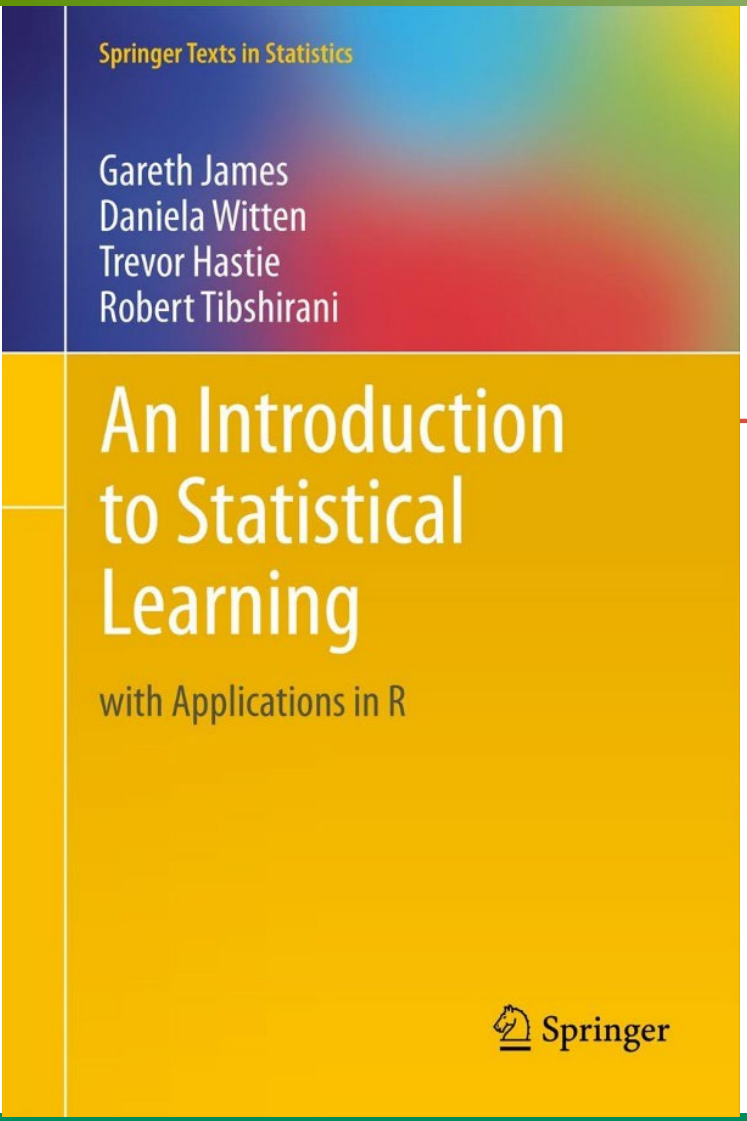


ANALYTICS

Week 8-9

Instructor:

Jeffrey Zheng

Examples Inspired By CAS, SOA, Coaching Actuaries ©

Key Formulas This Week

- Breusch-Pagan: Reject H_0 if $\frac{SSReg_{aux}}{2} > \chi^2_{\alpha,p}$.
- Standardized residual: $\frac{e_i}{s\sqrt{1-h_{ii}}}$
- Leverage: $h_{ii} = \boldsymbol{x}_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{x}_i$, $h_{avg} = \frac{p+1}{n}$
- SLR: $h_i = \frac{1}{n} + \frac{(x_i-\bar{x})^2}{\sum_{j=1}^{n}(x_j-\bar{x})^2}$
- Cook's Distance: $D_i = \left(\frac{e_i}{s\sqrt{1-h_{ii}}}\right)^2 \frac{h_{ii}}{(p+1)(1-h_{ii})}$
- Variance Inflation: $VIF_j = \frac{1}{1-R_j^2}, j = 1,2,\ldots p$

MLR Assumptions Recap

- Multiple linear regression model assumptions:
- 1. $Y_i = \beta_0 + \beta_1 x_{i1} + \cdots \beta_p x_{ip} + \varepsilon_i$ is the model
- 2. $\{x_{i1}, x_{i2}, \ldots, x_{ip}\}$ are **non-stochastic variables**.
- 3. $E[\varepsilon_i] = 0$
- 4. **Homoscedasticity**: $Var(\varepsilon_i) = \sigma^2$
- 5. $y_i's$ / $\varepsilon_i's$ are **independent**.
- 6. $y_i's$ / $\varepsilon_i's$ are **normally distributed.**
- 7. Predictors are **non-collinear**.

MLR Violations

- These mirror the regression assumptions.

- 1. **Mis-specified model**: model form used is incorrect
- 2. Residuals have non-zero averages
- 3. **Heteroscedasticity**: non-constant variance of residuals

MLR Violations 2

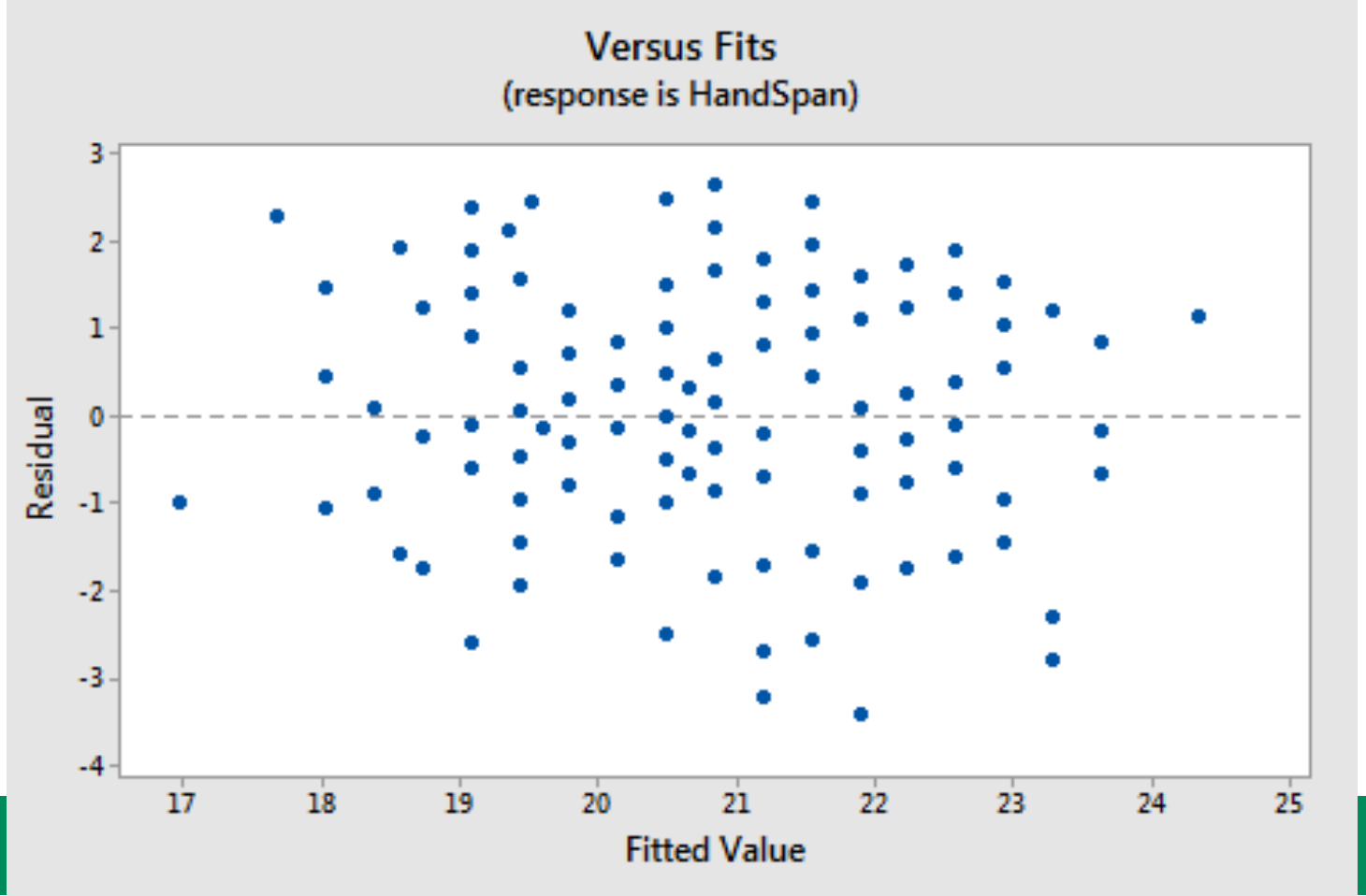
- 4. Residuals exhibit **serial correlation** : lack indep., relations w/ prior residuals
- 5. **Non-normality** : residuals non-normal

MLR Issues

- These are potential issues to investigate.
- **Outliers** – extreme residuals (y's)
- **High leverage points** – unusual predictor combination (x's)
- **Multi-collinearity** – perfect - viol. predictor nearly a linear combination of other predictors
- **High dimensionality** – too large p vs. n

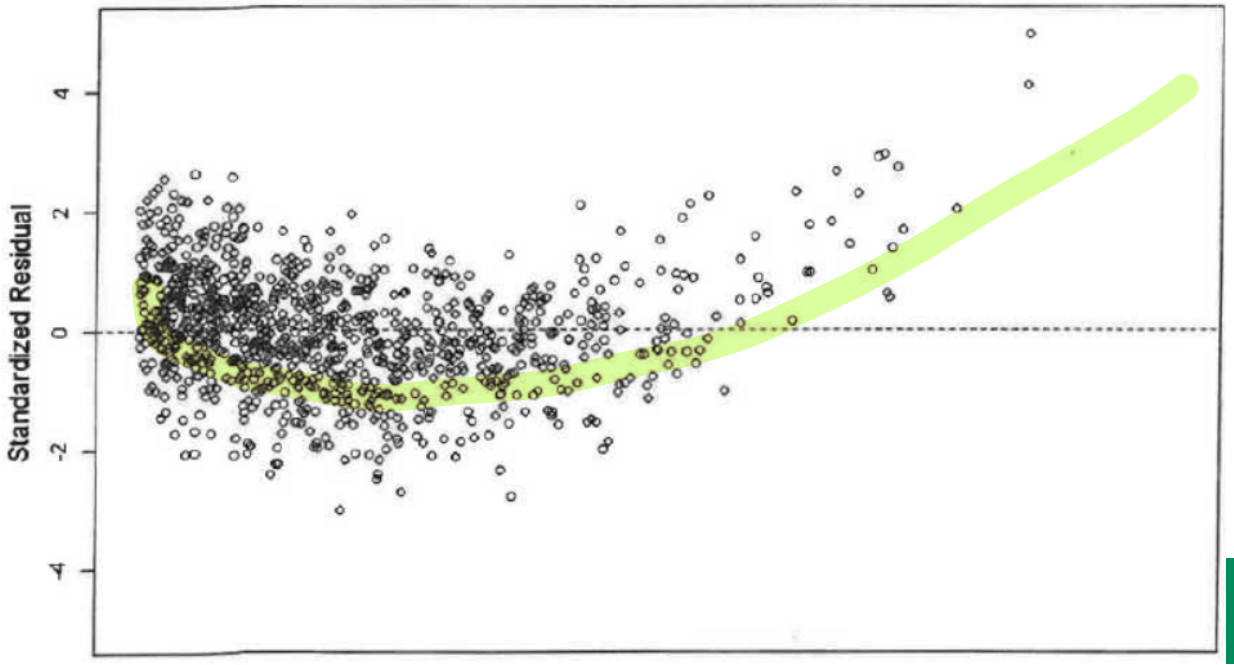
Good Residuals

- No discernable patterns, random scatter about 0 and average to 0, consistent spread



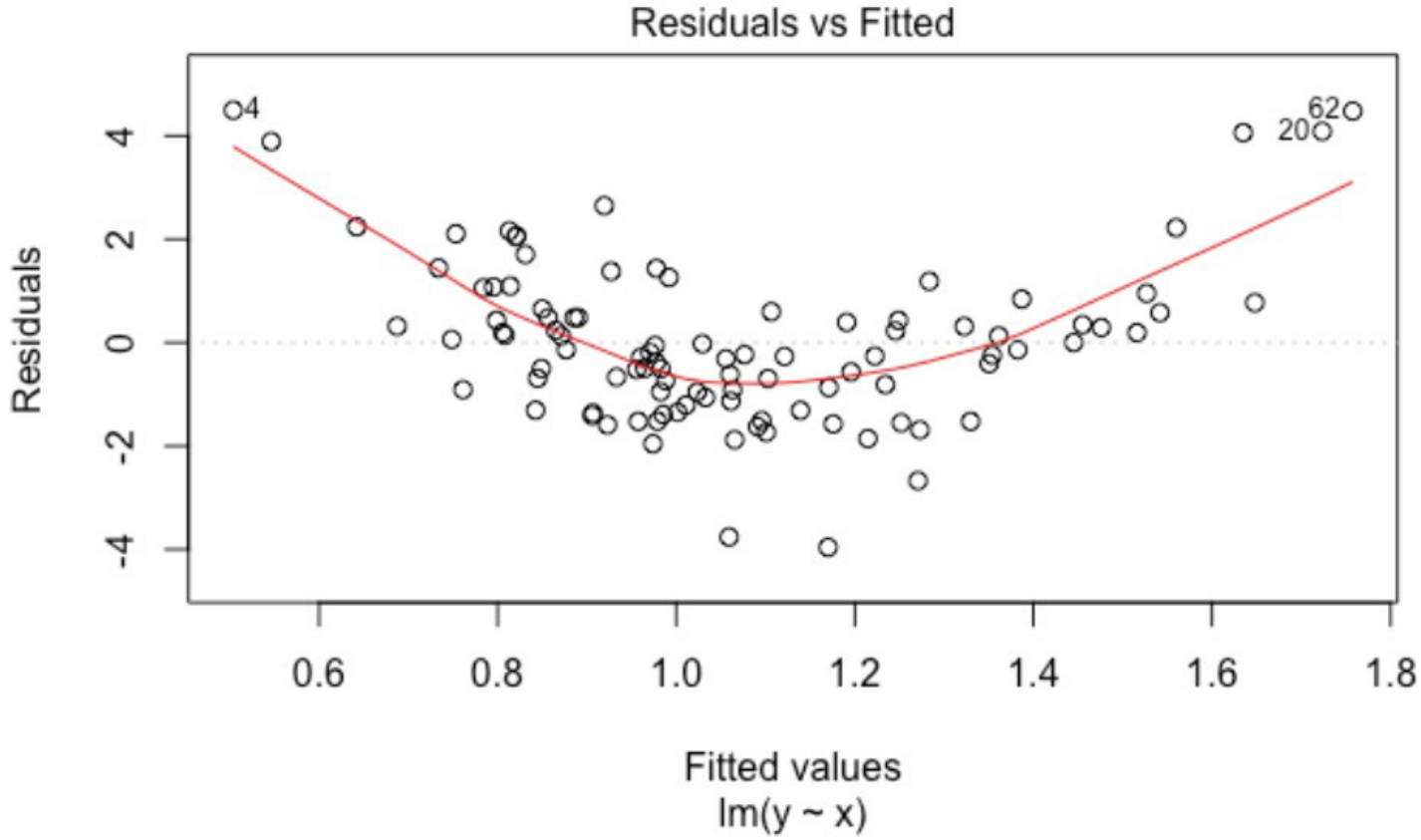
#1. Mis-Specified Model

- Residuals with a clear non-linear pattern may indicate the need to include additional terms.
- Underfitting Consequences: biased coef. estim., inflated MSE, se's, wider pred. intervals



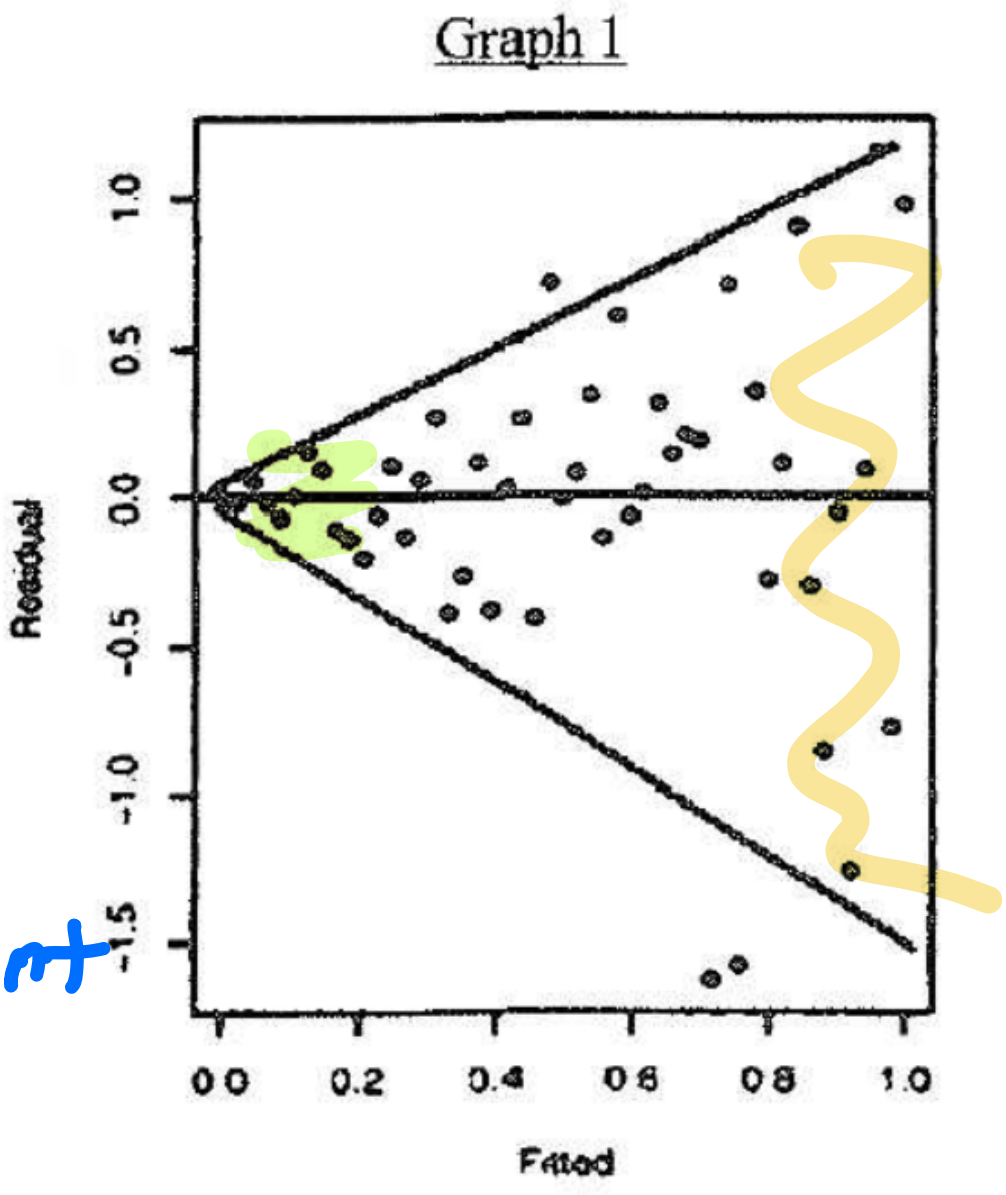
#2. Non-Zero Averages

- https://thestatsgeek.com/2020/03/23/the-mean-of-residuals-in-linear-regression-is-always-zero/
- Non-zero averages in ranges implies a pattern.



#3. Heteroscedasticity (CAS)

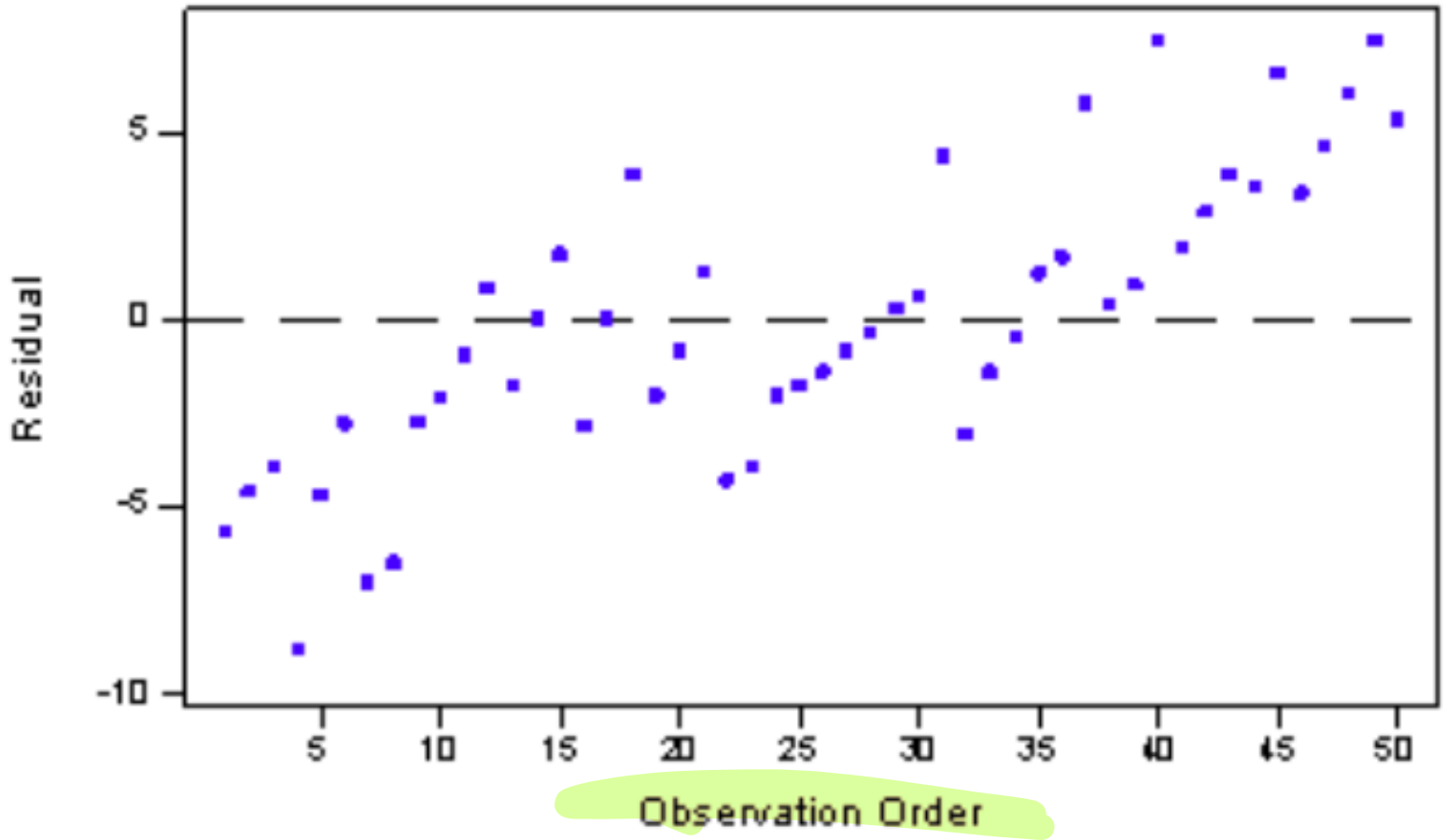
- Can ignore mild (still unbiased)
- Transforming Y may help
- Severe may require weighted least squares or hetero-scedasticity consistent standard errors



#4. Dependent Errors

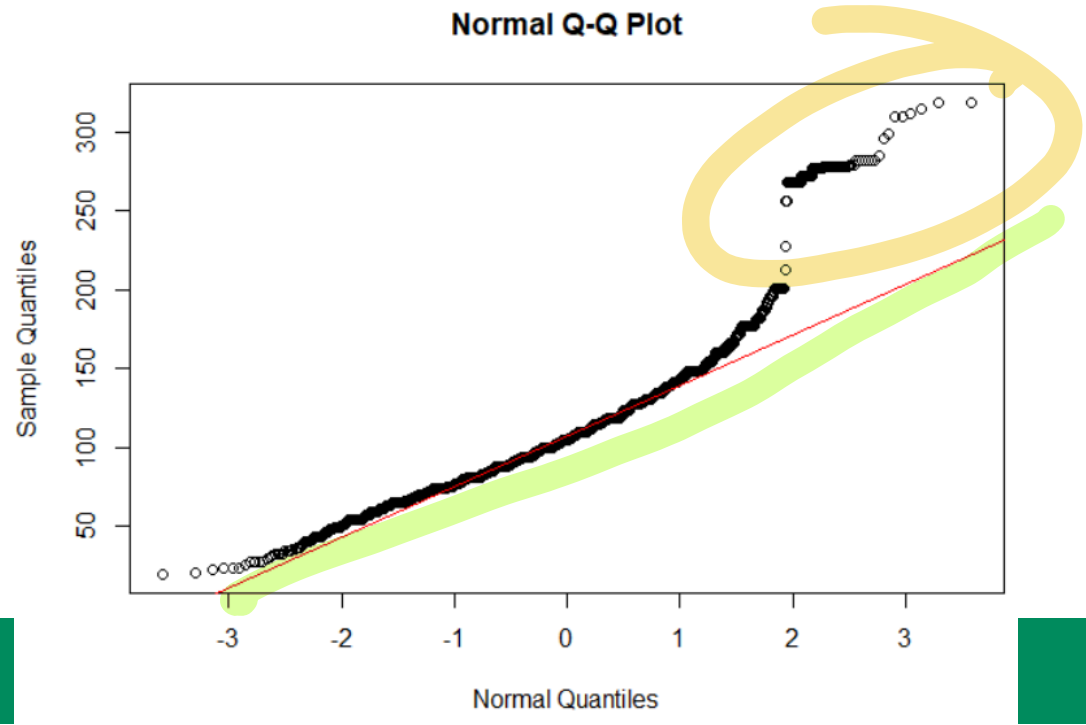
time series

- Serial correlation occurs when the residuals have a pattern across time



#5. Non-Normality

- Quantile-Quantile (**QQ Plots**): Measures data fit with reference model (typically normal)
- Follow the 45-degree reference line if normal
- Fixes: transformations, GLM



Residual Plots

- A plot of e against $\hat{y}$ detects non-linear relationships, non-zero averages, remaining trends, heteroscedasticity
- A plot of e against observation time detects serial correlation
- A QQ-plot of e detects non-normality

Breusch-Pagan Test

- The Breusch-Pagan Test is hypothesis test method to test if heteroscedasticity exists in a linear regression model, assuming normally distributed error terms.
- 1. Calculate squared standardized residuals from the model $r_i = \frac{e_i^2}{\hat{\sigma}^2}$ $\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} e_i^2}{n}$
- 2. Run a regression with r_i as the y_i's against the original predictors (save SSReg)

Breusch-Pagan Test 2

- H_0: Homoscedasticity in model
- H_a: Heteroscedasticity in model
- $LM = \frac{SSReg_{aux}}{2}$
  - $SSReg_{aux}$: Regression sum of squares for the second model
- Reject H_0 if $LM > \chi^2_{\alpha, p}$

1. B-P Illustration

- A linear regression model of Y (# of confusing trends) against X (# of books read) results in a mean-squared error of 25 and a F-statistic of 2 with 1 and 16 degrees of freedom.
- A linear regression model of the squared standardized residuals from the first model against X results in a $MSE = 5$ and $F_{1,16} = 0.5$.
- Use the Breusch-Pagan test at $\alpha = 5\%$ to test for heteroscedasticity.

$$0.5 = \frac{\frac{SSReg}{p}}{MSE} = \frac{SSReg_{aux}}{5}$$

$$SSReg_{aux} = 2.5$$

$$LM = \frac{2.5}{2} = 1.25$$

$$\chi^2_{1,.05} = 3.841$$

$1.25 < 3.841$, $\therefore$ fail to reject H_0

1. B-P Illustration 2

- $LM = \frac{SSReg_{aux}}{2}$
- $F_{p,n-(p+1)} = \frac{\frac{SSReg}{p}}{MSE}$

Heteroscedasticity Fixes

- Transform the response
- Use weighted least squares
- Adjust standard errors to heteroscedasticity – consistent standard errors
- FYI: $Var(\boldsymbol{b}) = \sum_{i=1}^{n} \boldsymbol{w}_i e_i^2 \boldsymbol{w}_i^T$ where $\boldsymbol{w}_i = (\boldsymbol{X}^T \boldsymbol{X})^{-1} x_i$

Weighted Least Squares

- In weighted least squares, the response becomes $\sqrt{w_i}y$ and the predictors $\sqrt{w_i}\boldsymbol{x}$.
- $Var(\varepsilon_i) = \frac{\sigma^2}{w_i}$
  - w_i: observation weight
- Goal: Minimize $\sum_{i=1}^{n} w_i(y_i - \hat{y}_i)^2$
- FYI: $b = (\boldsymbol{X}'\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{W}\boldsymbol{y}$

Leverage

high leverage and/or outlier driven

- **Influential points** have a strong effect on the overall model fit.

- High **leverage** points have an unusual ___

explanatory variable combination

(unusual x's)

Leverage 2

- $\hat{\boldsymbol{y}} = \boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y} = \boldsymbol{H}\boldsymbol{y}$ where $\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T$
  - $\boldsymbol{H}$: hat matrix (puts the hat on $\boldsymbol{y}$)

- $\widehat{Var}\,(e_i) = s^2(1 - h_{ii})$

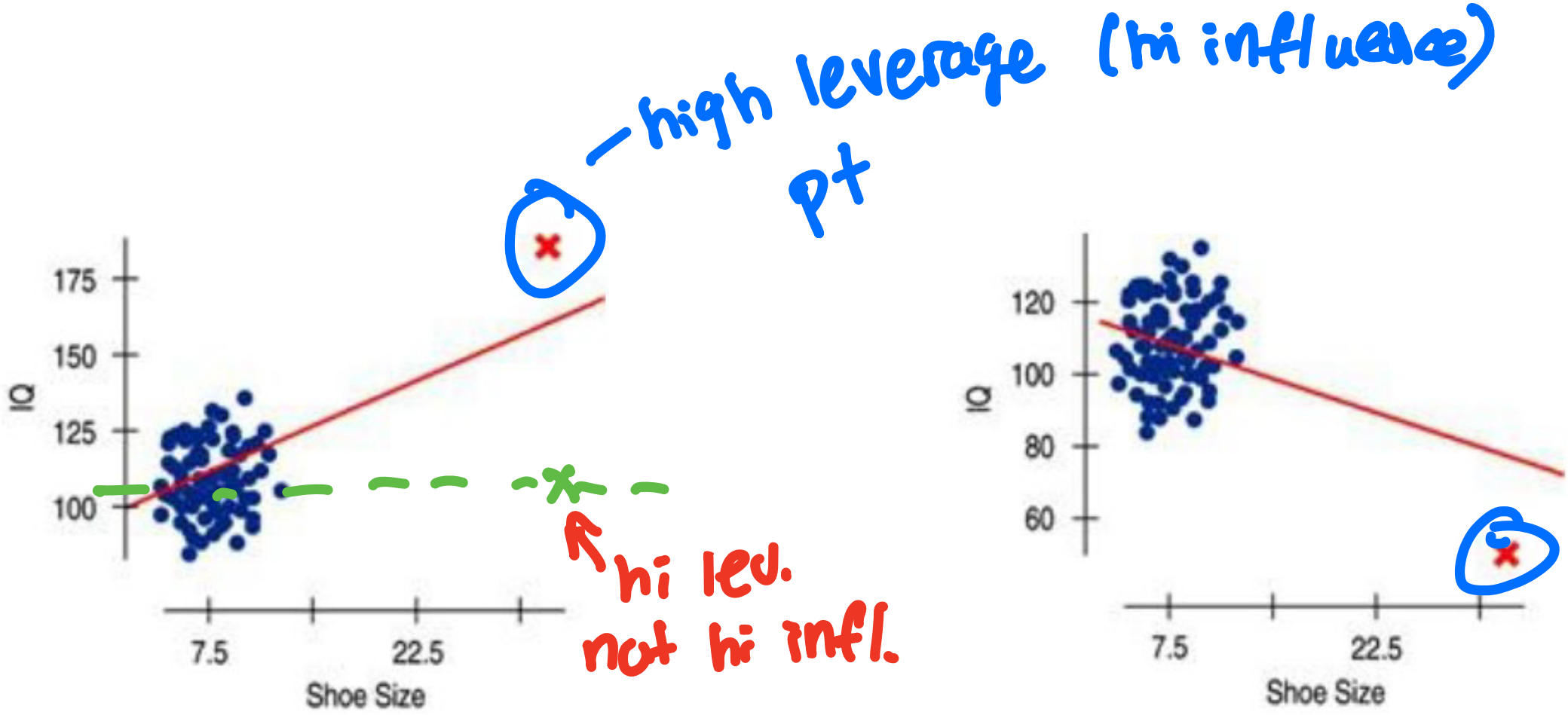
- $Cov(e_i, e_j) = -s^2\left(Cov_{ij} \text{ term from } \boldsymbol{H}\right)$

Leverage 3

- $\hat{y}_i = h_{i1}y_1 + h_{i2}y_2 + \cdots + h_{ii}y_i + \cdots h_{in}y_n$
  - h_{ij}: Row i, Column j entry from hat matrix $\boldsymbol{H}$
- Simple linear regression: $h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^{n}(x_j - \bar{x})^2}$
- $\frac{1}{n} \leq h_{ii} \leq 1$ always
- $h_{avg} = \frac{p+1}{n}$ with sum = p+1
  - p: # of explanatory predictors
  - n: # of data points
- High leverage point is typically defined as $h_{ii} > 3 \cdot \frac{p+1}{n}$

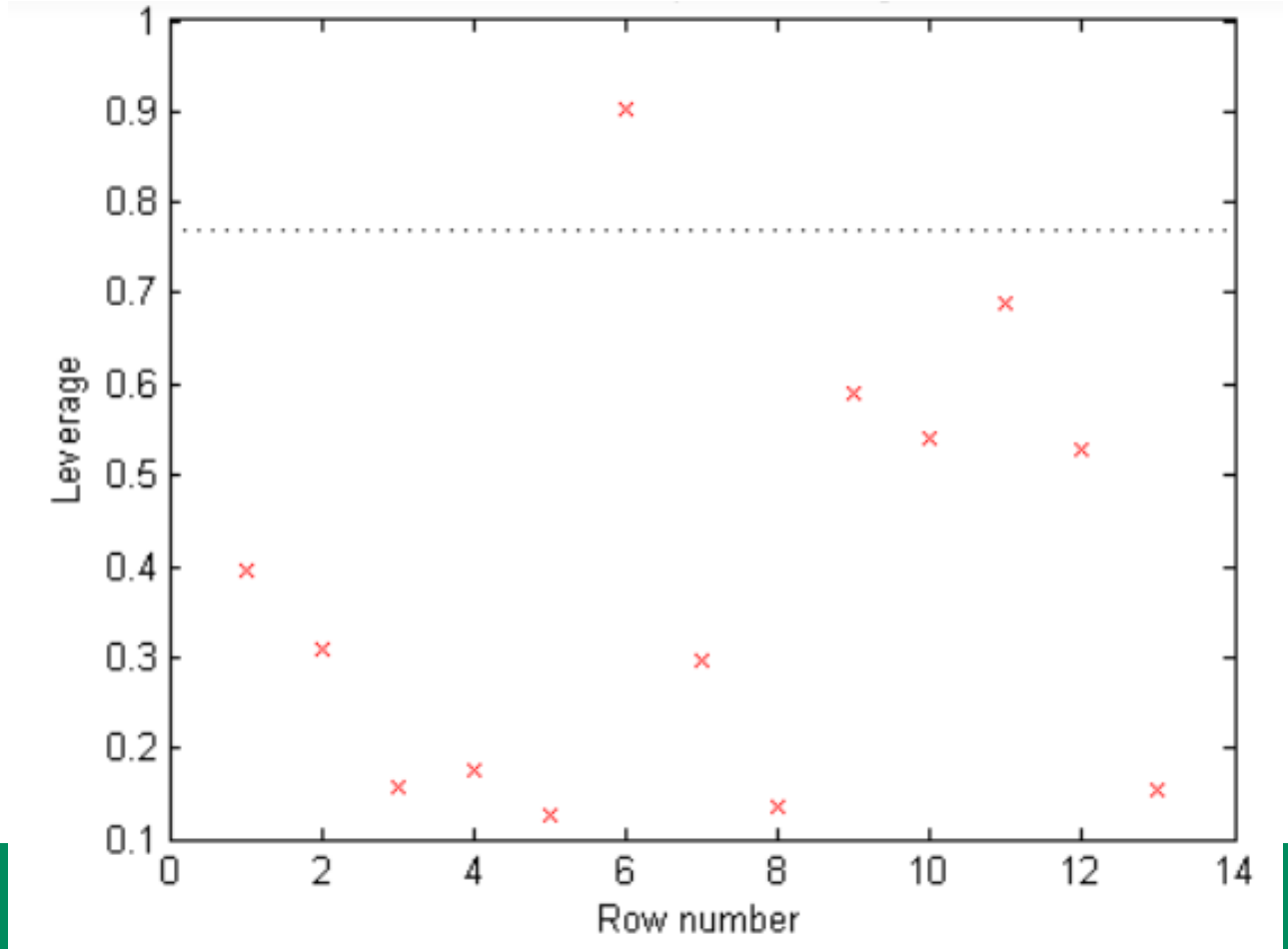
High Leverage Points

- https://slideplayer.com/slide/3599913/



Leverage Plot

- *Sample leverage plot*
- Residual, QQ plots do not detect high leverage points.



2. Leverage Illustration

- A basic linear regression model is fit to 100 observations. If $\sum_i x_i = 800$ and $\sum_i x_i^2 = 30{,}000$, calculate the leverage when $x = 5$.

$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^{n} (x_j - \bar{x})^2} = \frac{1}{100} + \frac{\left(5 - \frac{800}{100}\right)^2}{30{,}000 - \frac{800^2}{100}} = 0.01$$

2. Leverage Illustration 2

- $$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^{n}(x_j - \bar{x})^2} = \frac{1}{n} + \frac{\left(x_i - \frac{\sum_j x_j}{n}\right)^2}{\sum_j \left(x_j^2\right) - \frac{\left(\sum_j x_j\right)^2}{n}}$$

Leverage Handling

- Ways to handle high leverage points include:
- Include the observation & comment.

- Delete the observation if not representative, or present model with & without it

- Choose another explanatory variable to serve as a replacement/proxy

Leverage Handling 2

- Transform the explanatory variable.

- Use **robust estimation** techniques that reduce effect of influential points.

Standardized Residuals

- Standardizing residuals makes them dimension-less.
- Standardized residual = $\frac{e_i}{se(e_i)}$

- $e_i = y_i - \hat{y}_i$
- $se(e_i) = s\sqrt{1 - h_{ii}}$
  - s^2: Mean squared error (MSE)
  - h_{ii}: i^{th} diagonal element of hat matrix

Studentized Residuals

- **Studentized residuals** follow the same formula as the standardized residual, but the estimations are made with the ith observation excluded from the model.
- Studentized residuals are unit-less, follow a t-dist (assuming error normality), better detect large residuals

Outliers

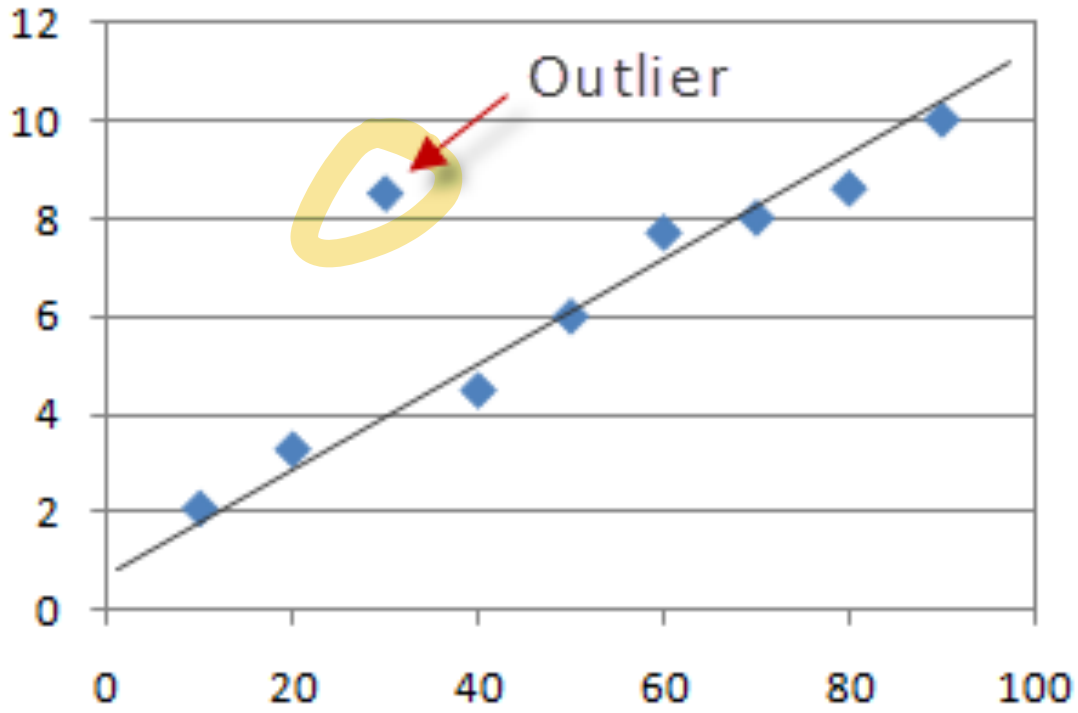
unusual y's

- Outliers are observations with a large residual, indicating it is not well fit by the model. They should be investigated.

- Typically, |standardized or studentized residuals| > 2 is considered an outlier.

Outlier Plot

- https://www.mathopenref.com/outlier.html



Handling Outliers

- Options for outliers:
- Include the observation but comment on it
- Delete the observation
- Create a binary variable to flag the outlier. If a cause drives the outlier, all points related to the cause can be flagged.

Concept Check #1

- Which statements are true?
- I. If residuals exhibit a predictable pattern from ordered observation to observation, transform the response to fix.
- II. The total leverage sums to the # of explanatory variables.
- III. Dependent MLR error terms lead to too high estimated standard errors.
- IV. If residuals are negative on the extremes and positive in the middle, adding a quadratic predictor may help.
- V. For simple linear regression with 10 observations, leverage above 0.6 is considered high.
- VI. High dimensional data makes R^2, the fitted model, and parameter confidence intervals all unreliable.
- VII. A log or square root transformation can make the variance of the residuals less constant.

F F F T T T F

+1

$3 \cdot \frac{2}{10} = .6$

goal is to

Example 1

- A basic linear regression model to predict # of daily tantrums by a child is fit to 4 observations.

- $\boldsymbol{X(X^TX)^{-1}X^T} = \begin{pmatrix} 0.6 & -0.2 & 0.1 & 0.3 \\ -0.2 & 0.3 & -0.1 & 0.5 \\ 0.1 & -0.1 & 0.4 & 0.01 \\ 0.3 & 0.5 & 0.01 & 0.7 \end{pmatrix}$ H

- $\boldsymbol{y} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 6 \end{pmatrix}$ and $\boldsymbol{X(X^TX)^{-1}X^Ty} = \begin{pmatrix} 2.2 \\ 3.3 \\ 1.46 \\ 7.04 \end{pmatrix}$ $= \hat{y}$

  H

- $s^2 = 2$

- Calculate the standardized residual of the second observation. Is it an outlier?

on our exams: H, β, $\hat{y}$ clearly given

$(X^TX)^{-1} \cdot s^2$ = Var-Cov matr.

$$r_{std} = \frac{y_i - \hat{y}_i}{s\sqrt{1-h_{ii}}}$$

$y_i - \hat{y}_i \to 3.3$

$s \to \sqrt{2}$

$h_{ii} \to .3$

$= -.254$

not outlier

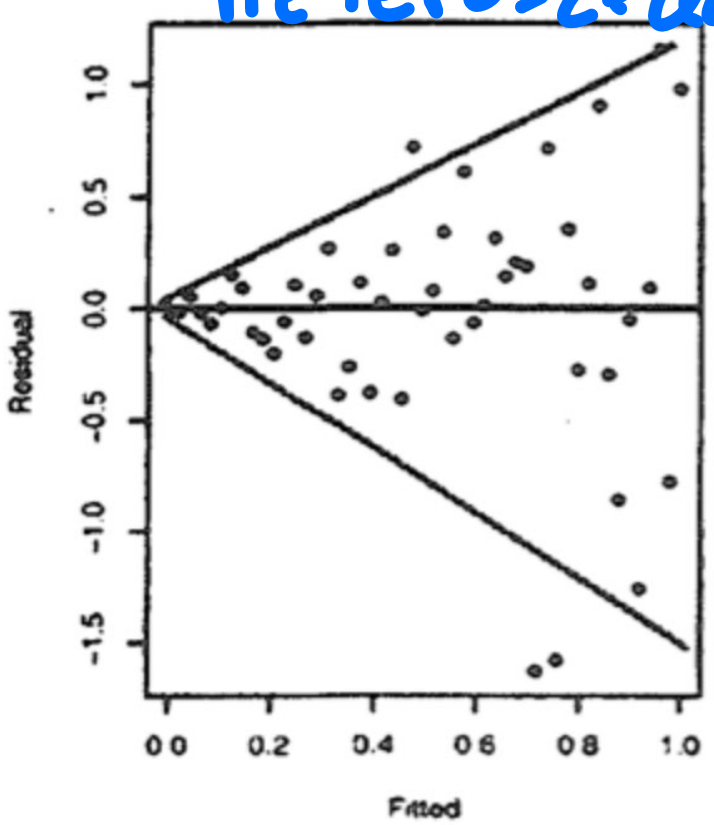
Example 1 Sol

- Standardized residual = $\frac{e_i}{se(e_i)}$
- $e_i = y_i - \hat{y}_i$, $se(e_i) = s\sqrt{1 - h_{ii}}$, $MSE = s^2$
- $\hat{y}_i = h_{i1}y_1 + h_{i2}y_2 + \cdots + h_{ii}y_i + \cdots h_{in}y_n$

Concept Check #2

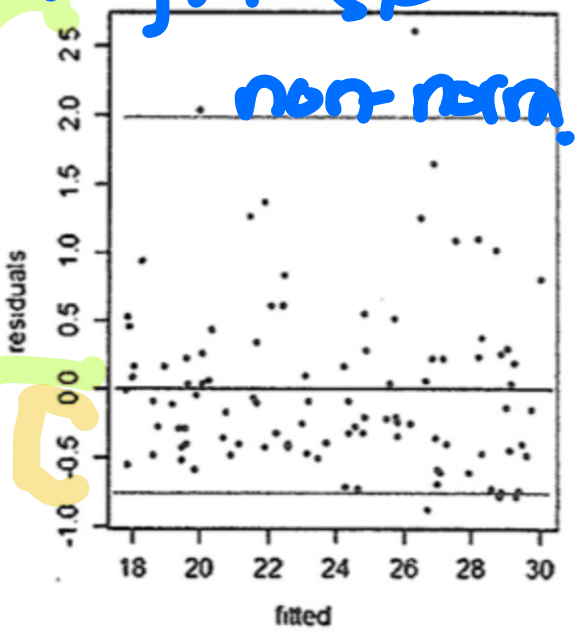
- For each graph, what MLR issues, if any, do they indicate?

heteroscedasticity

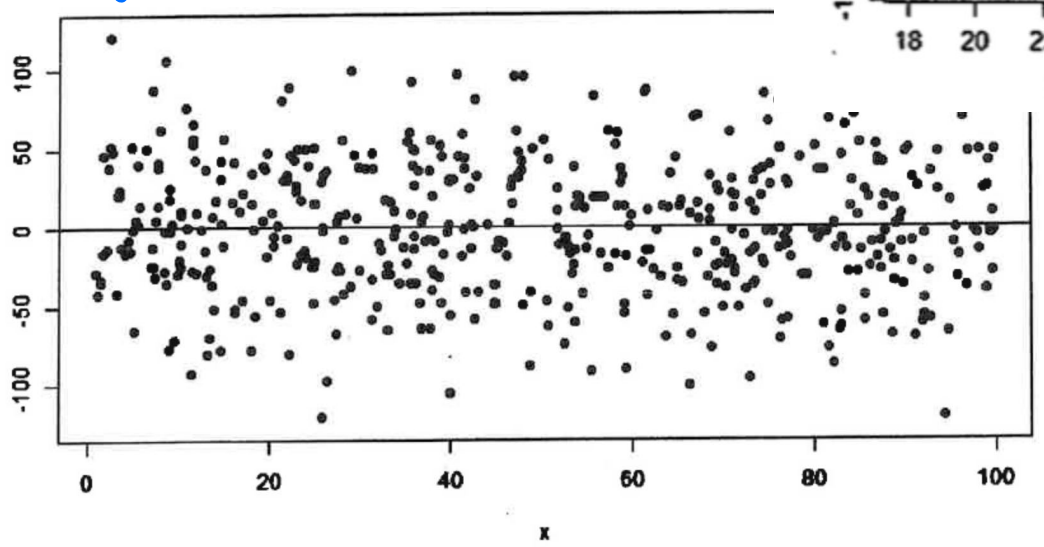


right-skew

non-norm.



no issues



Cook's Distance

- **Cook's distance** is a nice metric that quantifies the influence of a point (using both y's & x's)

- Average D_i is roughly $\frac{1}{n}$, with 3x the average considered high influence

- True comparison is to a F-statistic with $p + 1, n - (p + 1)$ degrees of freedom

Cook's Distance 2

- $D_i = \frac{\sum_{j=1}^{n}\left(\hat{y}_j - \hat{y}_{j(i)}\right)^2}{(p+1)s^2} = \left(\frac{e_i}{s\sqrt{1-h_{ii}}}\right)^2 \cdot \frac{h_{ii}}{(p+1)(1-h_{ii})}$

- $\hat{y}_{j(i)}$: prediction of jth observation with i^{th} observation left out of model fit
- p: # of explanatory predictors

3. Cook's D Illustration

- You fit a multiple linear regression model using 2 explanatory variables to predict the # of grumpy neighbors you have.
- Calculate leverage of the first observation and Cook's distance for the second observation.

$$0.125 = (-.5)^2 \cdot \frac{h_{1,1}}{(2+1)(1-h_{1,1})} \qquad h_{1,1} = .6$$

Obs. #	y_i	$\hat{y}_i$	h_{ii}	$\frac{e_i}{se(e_i)}$	D_i
1	10	12		-0.5 ✓	0.125
2	20	19	0.3		

$$D_2 = \left(\frac{20-19}{S\sqrt{1-.3}} \right)^2 \cdot \frac{.3}{(2+1)(1-.3)} = .0051$$

$$-.5 = \frac{10-12}{S\sqrt{1-.6}} \rightarrow S = 6.3246$$

3. Cook's D Illustration 2

- $D_i = \left(\frac{e_i}{s\sqrt{1-h_{ii}}}\right)^2 \frac{h_{ii}}{(p+1)(1-h_{ii})}$
- $e_i = y_i - \hat{y}_i,\ se(e_i) = s\sqrt{1-h_{ii}}$

Example 2 Data

- The model $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ is fit to the data:
- $(X^T X)^{-1} X^T y = \begin{bmatrix} 5.02 \\ -0.22 \\ -0.59 \end{bmatrix}$ $\hat{\beta}$
- $X(X^T X)^{-1} X^T y = \begin{bmatrix} 2.55 \\ 1.26 \\ \text{Missing} \\ 3.195 \\ 4.32 \end{bmatrix}$; $y = \begin{bmatrix} \text{Missing} \\ 2 \\ 3 \\ \text{Missing} \\ \text{Missing} \end{bmatrix}$ $\hat{y}$
- $X(X^T X)^{-1} X^T = \begin{bmatrix} 0.35 & 0.23 & -0.03 & 0.42 & 0.03 \\ 0.23 & 0.87 & 0.16 & -0.1 & -0.16 \\ -0.03 & 0.16 & 0.55 & -0.13 & 0.45 \\ 0.42 & -0.1 & -0.13 & 0.68 & 0.13 \\ 0.03 & -0.16 & 0.45 & 0.13 & 0.55 \end{bmatrix}$ H

(B) Hi lev std = $\frac{3(2+1)}{5} = 1.8$

no hi lev. pts.

(C) $.2935 = \left(\frac{e_1}{s\sqrt{1-.35}}\right)^2 \cdot \frac{.35}{(2+1)(1-.35)}$

$s = \sqrt{\frac{4.52}{5-(2+1)}} = 1.503$

$e_1 = -1.55$

(D) $r_{std,12} = \frac{2-1.26}{1.503\sqrt{1-.87}} = 1.365$

(E) $\hat{y}_3 = 5.02 - .22(.75) - .59(2)$
$= 3.675$

$$e_3 = 3 - 3.675 = -.675$$

$$D_3 = \left(\frac{-.675}{1.503\sqrt{1-.55}} \right)^2 \cdot \frac{0.55}{(2+1)(1-.55)}$$

$$= .1825$$

(F) $$SSE = \sum_i e_i^2$$

$$0 = \sum_i e_i$$

$e_4 = .819$, $e_5 = .666$

Example 2 Qs

- A. Identify all the matrices.
- B. How many are high leverage points according to the rule of thumb?
- C. If the sum of squared residuals is 4.52 and the Cook's distance for the first point is 0.2935, what is the first residual (if negative)?
- D. What is the second standardized residual?
- E. If $x_{1,3} = 0.75, x_{2,3} = 2$ for the third observation, what is the third Cook's distance?
- F. (Bonus) What are the residuals for the 4th and 5th observations if $e_4 > e_5$?

Example 2 Sol

- $D_i = \left(\frac{e_i}{s\sqrt{1-h_{ii}}}\right)^2 \frac{h_{ii}}{(p+1)(1-h_{ii})}$
- $MSE = s^2 = \frac{SSE}{n-(p+1)}$
- High Leverage: $\frac{3(p+1)}{n}$
- $e_{std,i} = \frac{e_i}{s\sqrt{1-h_{ii}}}$
- $e_i = y_i - \hat{y}_i$

Collinearity

- **Collinearity**/multicollinearity occurs when an explanatory variable is, or almost is, a linear combination of other available explanatory variables.
- If perfect collinearity exists, if all those variables are used, there are an infinite # of possible fits
- Near collinearity does not prevent model fitting, but leads to high standard errors of β's

Collinearity 2

- Collinearity relates to predictor-predictor relationships, not response - predictor relations.

  ie. won't dir increase s^2

- The F-test is not affected by multi-collinearity.

Collinearity Detection

- A correlation matrix of predictors or pairwise scatterplots can help detect collinearity (but can miss collinearity among 3+ predictors).

- Residual, leverage, Cook's Distance plots will not detect collinearity.

VIF

- The **variance inflation factor** (VIF) is a measure of collinearity of an explanatory variable. **Tolerance** is the inverse of VIF.
- $VIF_j = \frac{1}{1 - R_j^2}$ $j = 1, 2, \ldots, p$
  - VIF_j: VIF for j^{th} explanatory variable
  - R_j^2: Coefficient of determination when x_j is used as the response against all other $x's$
- $VIF_j \geq 10$ indicates severe collinearity
- $5 < VIF_j < 10$ indicates mild collinearity

Special Variable Terms

(perpendicular)

- **Orthogonal** predictors are completely ____
  uncorrelated (lack of collinearity)

- A **suppressor variable** is an explanatory variable ____
  not very correlated w/ y but increases the importance of other explanatory variables when included. (should incl even w/ some collinearity)

Concept Check #3

- Which statements are true?

T
- I. The leverage for each observation of the linear model is between $\frac{1}{n}$ and 1.

T
- II. If the regression of X_1 on X_2 yields $R^2 = 0$, the two variables are orthogonal.

F
- III. The VIF for an uncorrelated variable is ~~0~~ 1.

F
- IV. Heteroscedasticity leads to an unreliable VIF.

T
- V. Under perfect multi-collinearity, regression coefficients cannot be uniquely estimated.

F
- VI. Dependent MLR error terms lead to multi-collinearity.

T
- VII. If two variables are non-significant using t-tests, but the F-tests suggests at least one is significant, the variables could be negatively correlated and the model may be experiencing collinearity.

F
- VIII. Collinearity always implies one variable is redundant in the presence of another variable.

4. VIF Illustration

- In a multiple linear regression model for predicting alien life on other planets using 2 explanatory variables $\{x_1, x_2\}$, $R_1^2 = \frac{3125 - 625}{3125} = .8$
- $\sum(x_{i1} - \bar{x}_1)^2 = 3{,}125$ "SST"
- ~~$\sum(y - \bar{y})^2 = 5{,}900$~~
- For a regression of ~~$y \sim x_1 + x_2$~~, the error sum of squares is ~~500~~.
- For a regression of $x_1 \sim x_2$, the error sum of squares is 625. = SSE
- Calculate the variance inflation factor for x_1, x_2.

$VIF_1 = \frac{1}{1 - .8} = 5 = VIF_2$

4. VIF Illustration 2

- $VIF_j = \frac{1}{1-R_j^2}$
- $R^2 = \frac{SSR}{SST}$ and $SST = SSE + SSR$
- $\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$

Collinearity Handling

- Options to deal with high collinearity:
- Standardize the variables
- Accept and discuss its implications.
- Transform into orthogonal predictors

Collinearity Handling 2

- Replace a collinear variable with a proxy
- Remove one or more variables.
- Combine predictors into one

Collinearity & Leverage

- Both collinearity and leverage are used for model criticism and based only on the explanatory variables
- Both naturally exist and do not inherently make a deficient model, but rather are useful for model improvement.
- Collinearity can be masked or conversely created by high leverage points.

Issues Summary

Cause	Effect
Mis-specified model	Unreliable $b's$, MSE, $se's$, inference metrics
Heteroscedasticity	Unreliable $b's$, MSE
Dependent errors	Unreliable $b's$, $se's$ too low
Non-normality	Unreliable inference metrics
Outliers	Unreliable MSE
High leverage points	Unreliable $b's$
Multi-collinearity	Unreliable $b's$, $se's$ too high
High dimensionality	Unreliable $b's$, MSE

Concept Check #4

- Which of the following actions can be taken to address multi-collinearity?
- ✓ I. Do nothing
- ✓ II. Use orthogonal predictors
- ✗ III. Transform the response variable
- ✓ IV. Remove predictors
- ✓ V. Standardize predictors
- ✓ VI. Combine predictors

Example 3

- Given the following info, determine which predictors might be good to drop from the model. Predictors should not be removed from the model solely based on the variance inflation factor.

Predictor	VIF	95% Confidence Interval
x_1 drop?	3	$(-3.1, 0.5)$
x_2 drop? (1st)	7	$(-7.2, 5.9)$
x_3 keep?	8	$(-4.4, 0.1)$

Example 4 (CAS)

- Two different data sets were used to construct the four regression models below.
- A. How collinear is dataset A?
- B. What range of SSE would make dataset B mildly but not severely collinear?

Data Set	Model	Dependent Variables	Independent Variables	SST	SSE
A	1	Y_A	X_{A1}, X_{A2}	36,000	2,800
A	2	X_{A1}	X_{A2}	9,000	600
B	3	Y_B	X_{B1}, X_{B2}	30,000	?
B	4	X_{B1}	X_{B2}	80,000	??

$$VIF_A = \frac{1}{1 - \frac{9000-600}{9000}} = 15$$

severely collinear

$$5 < \frac{1}{1 - \frac{80{,}000-SSE}{80{,}000}} < 10$$

$$8000 < SSE < 16{,}000$$

Example 4 Sol

- $VIF_j = \frac{1}{1-R_j^2}$
- $R^2 = \frac{SSR}{SST}$ and $SST = SSE + SSR$

HW Note

- Some Model Selection concepts are mixed into part of Model Assumption Qs.

- Validation/hold-out set validation fits 1 model (computationally cheapest).
- k-fold cross-validation fits k models.
- LOOCV is k-fold with $k = n$ models.

Summary Check

- Linear Model Violations (Mis-specified Model, Non-Zero Residual Averages, Heteroscedasticity, Dependent Errors, Non-Normality) and any potential fixes
- Linear Model Issues (Outliers, High Leverage, Collinearity, High Dimensions) and any potential fixes
- Formulas and properties for Breusch-Pagan, leverage, standardized residuals, Cook's distance, VIF